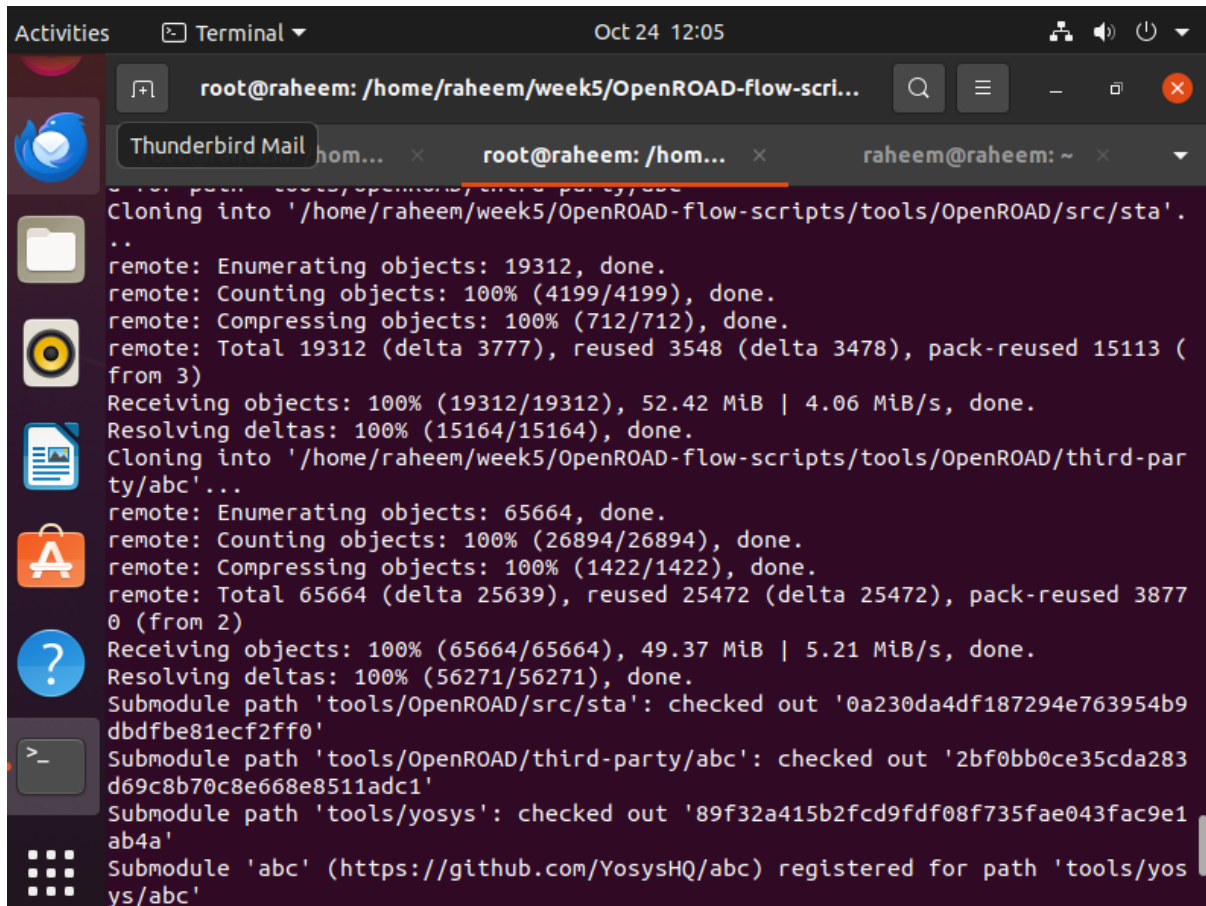


OpenROAD is an open-source, fully automated **RTL-to-GDSII** toolchain for **digital IC (Integrated Circuit) design**. It offers an end-to-end physical design flow — from register-transfer level (RTL) synthesis to final layout (GDSII) output — supporting synthesis, floorplanning, placement, clock tree synthesis, routing, and layout generation. Its modular, scriptable, and GUI-supported environment makes it suitable for both **academic research** and **industrial prototyping**.

A screenshot of a Linux terminal window. The title bar shows 'Activities', 'Terminal', and the date 'Oct 24 12:05'. The terminal prompt is 'root@raheem: /home/raheem/week5/OpenROAD-flow-scri...'. The output shows the cloning of the OpenROAD repository and its submodules. The first clone is for the main repository, and the second is for the 'third-party' submodule. Both are completed successfully with progress bars and object counts. The submodules are also checked out and registered.

```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/src/sta'.
..
remote: Enumerating objects: 19312, done.
remote: Counting objects: 100% (4199/4199), done.
remote: Compressing objects: 100% (712/712), done.
remote: Total 19312 (delta 3777), reused 3548 (delta 3478), pack-reused 15113 (
from 3)
Receiving objects: 100% (19312/19312), 52.42 MiB | 4.06 MiB/s, done.
Resolving deltas: 100% (15164/15164), done.
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD/third-par
ty/abc'...
remote: Enumerating objects: 65664, done.
remote: Counting objects: 100% (26894/26894), done.
remote: Compressing objects: 100% (1422/1422), done.
remote: Total 65664 (delta 25639), reused 25472 (delta 25472), pack-reused 3877
0 (from 2)
Receiving objects: 100% (65664/65664), 49.37 MiB | 5.21 MiB/s, done.
Resolving deltas: 100% (56271/56271), done.
Submodule path 'tools/OpenROAD/src/sta': checked out '0a230da4df187294e763954b9
dbdfbe81ecf2ff0'
Submodule path 'tools/OpenROAD/third-party/abc': checked out '2bf0bb0ce35cda283
d69c8b70c8e668e8511adc1'
Submodule path 'tools/yosys': checked out '89f32a415b2fcd9fdf08f735fae043fac9e1
ab4a'
Submodule 'abc' (https://github.com/YosysHQ/abc) registered for path 'tools/yos
ys/abc'
```

Step 1:: Install Prerequisites

Update your system and install core build tools:

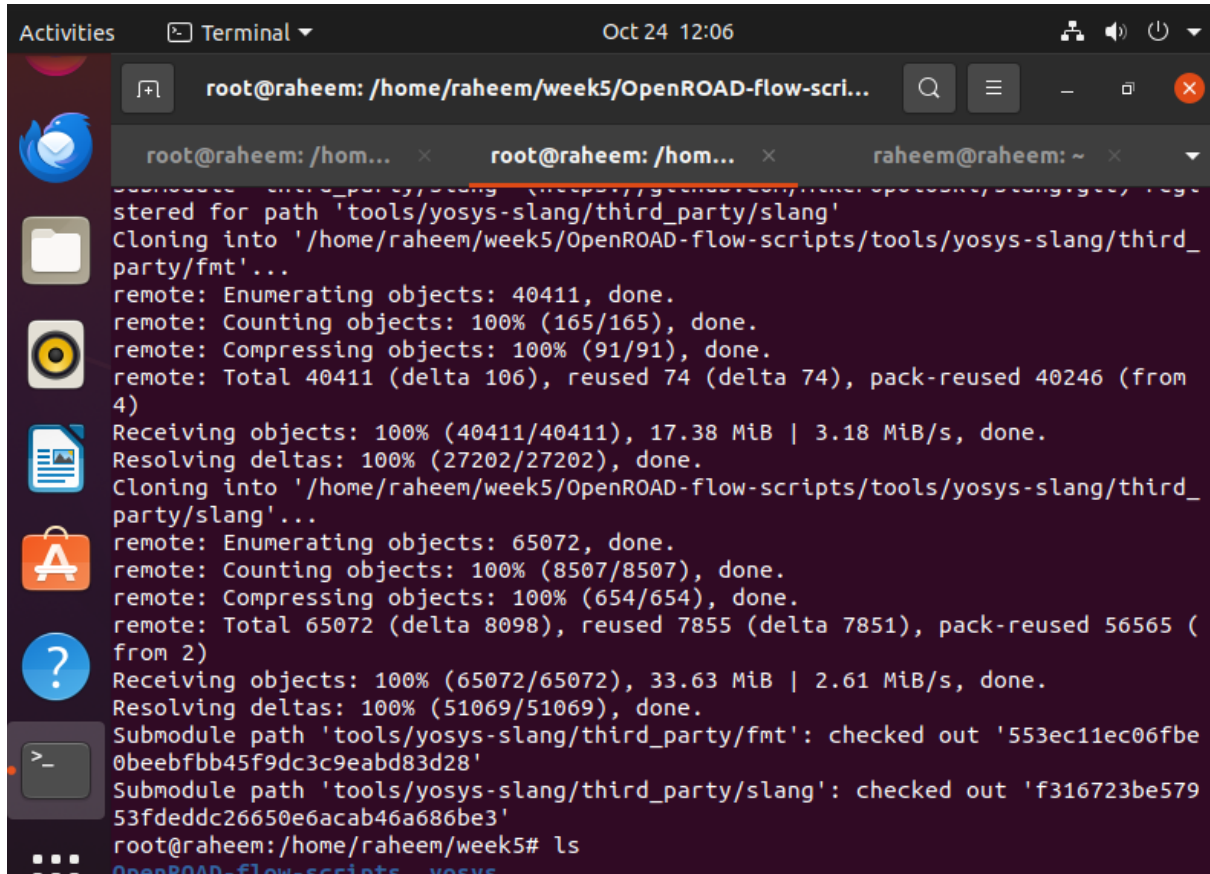
```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install -y build-essential cmake clang g++ gcc git
python3 python3-dev \
```

```
libboost-all-dev libtcl tcl-dev tcllib libreadline-dev zlib1g-dev
flex bison \
```

swig libpcre3-dev qtbase5-dev liblemmon-dev libspdlog-dev
libeigen3-dev libffi-dev \

pkg-config libjson-c-dev libzstd-dev



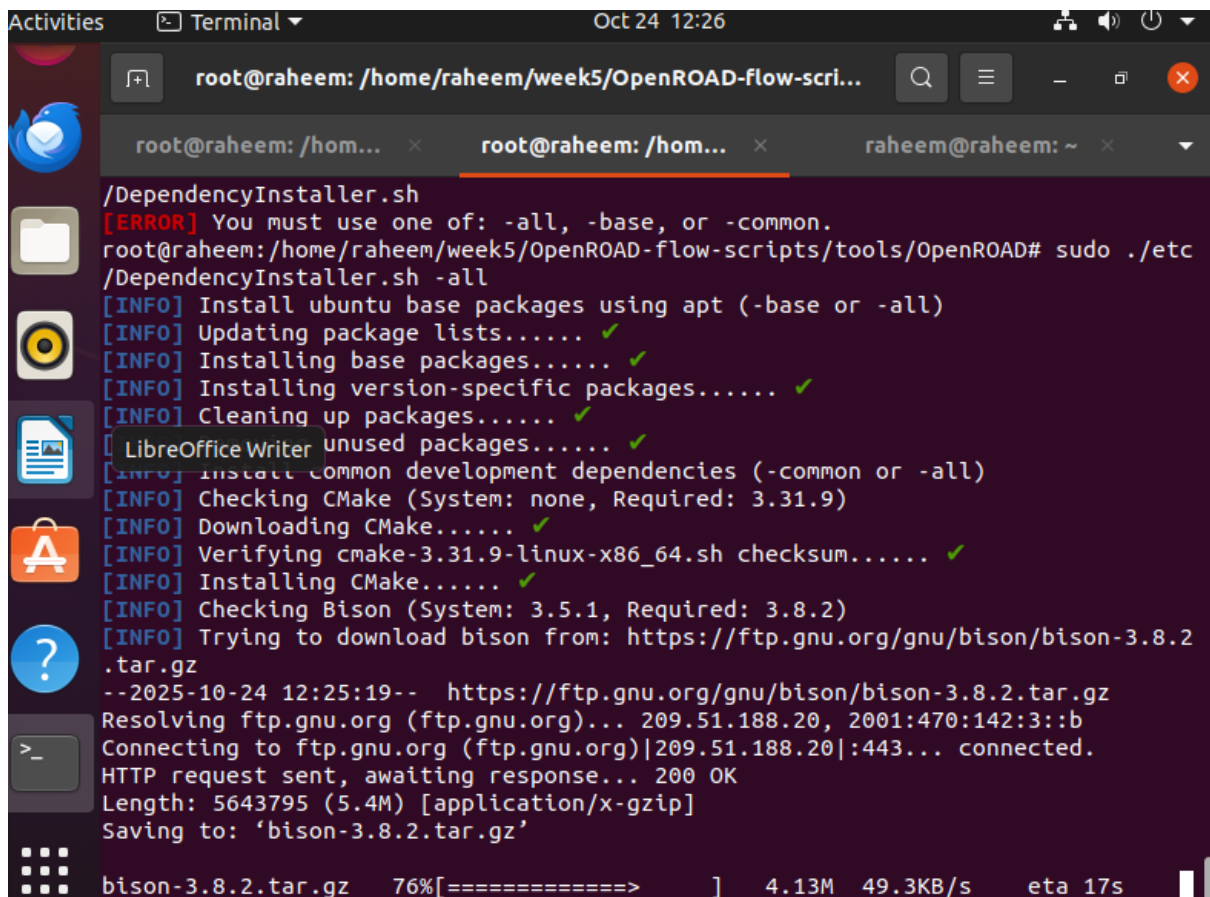
```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@raheem: /hom... x root@raheem: /hom... x raheem@raheem: ~ x
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/yosys-slang/third_
party/fmt'...
remote: Enumerating objects: 40411, done.
remote: Counting objects: 100% (165/165), done.
remote: Compressing objects: 100% (91/91), done.
remote: Total 40411 (delta 106), reused 74 (delta 74), pack-reused 40246 (from
4)
Receiving objects: 100% (40411/40411), 17.38 MiB | 3.18 MiB/s, done.
Resolving deltas: 100% (27202/27202), done.
Cloning into '/home/raheem/week5/OpenROAD-flow-scripts/tools/yosys-slang/third_
party/slang'...
remote: Enumerating objects: 65072, done.
remote: Counting objects: 100% (8507/8507), done.
remote: Compressing objects: 100% (654/654), done.
remote: Total 65072 (delta 8098), reused 7855 (delta 7851), pack-reused 56565 (
from 2)
Receiving objects: 100% (65072/65072), 33.63 MiB | 2.61 MiB/s, done.
Resolving deltas: 100% (51069/51069), done.
Submodule path 'tools/yosys-slang/third_party/fmt': checked out '553ec11ec06fbc
0beebfbb45f9dc3c9eabd83d28'
Submodule path 'tools/yosys-slang/third_party/slang': checked out 'f316723be579
53fdeddc26650e6acab46a686be3'
root@raheem: /home/raheem/week5# ls
OpenROAD-flow-scripts yosys
```

Step 2: Clone the Repositories

Install OpenROAD-flow scripts (wrapper for Yosys, OpenROAD, etc.)

```
git clone --recursive https://github.com/The-OpenROAD-Project/OpenROAD-flow-scripts.git
```

```
cd OpenROAD-flow-scripts
```



The image shows a terminal window titled "Terminal" with the date and time "Oct 24 12:26". The terminal is running a script located at `/home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD#`. The user has entered the command `sudo ./etc/DependencyInstaller.sh -all`. The script output shows various installation steps for Ubuntu base packages, development dependencies, CMake, and Bison. The progress bar at the bottom indicates that the download of `bison-3.8.2.tar.gz` is 76% complete.

```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@raheem: /hom... x root@raheem: /hom... x raheem@raheem: ~ x
/DependencyInstaller.sh
[ERROR] You must use one of: -all, -base, or -common.
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD# sudo ./etc
/DependencyInstaller.sh -all
[INFO] Install ubuntu base packages using apt (-base or -all)
[INFO] Updating package lists..... ✓
[INFO] Installing base packages..... ✓
[INFO] Installing version-specific packages..... ✓
[INFO] Cleaning up packages..... ✓
[INFO] LibreOffice Writer unused packages..... ✓
[INFO] Install common development dependencies (-common or -all)
[INFO] Checking CMake (System: none, Required: 3.31.9)
[INFO] Downloading CMake..... ✓
[INFO] Verifying cmake-3.31.9-linux-x86_64.sh checksum..... ✓
[INFO] Installing CMake..... ✓
[INFO] Checking Bison (System: 3.5.1, Required: 3.8.2)
[INFO] Trying to download bison from: https://ftp.gnu.org/gnu/bison/bison-3.8.2
.tar.gz
--2025-10-24 12:25:19-- https://ftp.gnu.org/gnu/bison/bison-3.8.2.tar.gz
Resolving ftp.gnu.org (ftp.gnu.org)... 209.51.188.20, 2001:470:142:3::b
Connecting to ftp.gnu.org (ftp.gnu.org)|209.51.188.20|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 5643795 (5.4M) [application/x-gzip]
Saving to: 'bison-3.8.2.tar.gz'
bison-3.8.2.tar.gz 76%[=====] 4.13M 49.3KB/s eta 17s
```

Step 3: Install and Build Dependencies

`sudo ./etc/DependencyInstaller.sh`

```
Activities Terminal Oct 24 12:27
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...

Saving to: 'bison-3.8.2.tar.gz'
bison-3.8.2.tar.gz 100%[=====] 5.38M 101KB/s in 79s
2025-10-24 12:26:39 (69.9 KB/s) - 'bison-3.8.2.tar.gz' saved [5643795/5643795]

[INFO] Verifying bison-3.8.2.tar.gz checksum..... ✓
[INFO] Extracting Bison..... ✓
[INFO] Configuring Bison..... ✓
[INFO] Building and installing Bison..... ✓
[INFO] Checking Flex (System: 2.6.4, Required: 2.6.4)
[INFO] Checking PCRE (System: none, Required: 10.42)
[INFO] Downloading PCRE..... ✓
[INFO] Verifying pcre2-10.42.tar.gz checksum..... ✓
[INFO] Extracting PCRE..... ✓
[INFO] Configuring PCRE..... ✓
[INFO] Building PCRE..... ✓
[INFO] Installing PCRE..... ✓
[INFO] Checking SWIG (System: none, Required: 4.3.0)
[INFO] Downloading SWIG..... ✓
[INFO] Verifying v4.3.0.tar.gz checksum..... ✓
[INFO] Extracting SWIG..... ✓
[INFO] Checking PCRE (System: 10.42, Required: 10.42)
[INFO] Generating SWIG configure script..... ✓
[INFO] Configuring SWIG..... ✓
```

```
Activities Terminal Oct 24 12:32
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...

[INFO] Checking or-tools (System: none, Required: 9.11.4210)
[INFO] Downloading or-tools..... ✓
[INFO] Extracting or-tools..... ✓

[INFO] Installation Summary
=====
====
Package | System Version | Required Version | Status
-----|-----|-----|-----
CMake | none | 3.31.9 | installed
Bison | 3.5.1 | 3.8.2 | installed
Flex | 2.6.4 | 2.6.4 | skipped
PCRE | none | 10.42 | installed
S Ubuntu Software | none | 4.3.0 | installed
Boost | none | 1.86 | installed
Eigen | none | 3.4 | installed
CUDD | none | 3.0.0 | installed
CUSP | none | any | installed
Lemon | none | 1.3.1 | installed
spdlog | none | 1.15.0 | installed
gtest | none | 1.17.0 | installed
Abseil | none | 20240722 | installed
or-tools | none | 9.11.4210 | installed
=====
root@raheem: /home/raheem/week5/OpenROAD-flow-scripts/tools/OpenROAD#
```

```
Activities Terminal Oct 24 14:08
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@raheem: /hom... x root@raheem: /hom... x raheem@raheem: ~ x
(OSS CAD Suite) root@raheem: /home/raheem/week5/OpenROAD-flow-scripts# ./etc/Dep
endencyInstaller.sh -all
[INFO] Install ubuntu base packages using apt (-base or -all)
[INFO] Updating package lists..... ✓
[INFO] Installing base packages..... ✓
[INFO] Installing version-specific packages..... ✓
[INFO] Cleaning up packages..... ✓
[INFO] Removing unused packages..... ✓
[INFO] Install common development dependencies (-common or -all)
[INFO] Checking CMake (System: 3.31.9, Required: 3.31.9)
[INFO] Checking Bison (System: 3.8.2, Required: 3.8.2)
[INFO] Checking Flex (System: 2.6.4, Required: 2.6.4)
[INFO] Checking PCRE (System: 10.42, Required: 10.42)
[INFO] Checking SWIG (System: 4.3.0, Required: 4.3.0)
[INFO] Checking Boost (System: 1.86, Required: 1.86)
[INFO] Checking Eigen (System: 3.4, Required: 3.4)
[INFO] Checking CUDD (Required: 3.0.0)
[INFO] Checking CUSP
[INFO] Checking Lemon (System: 1.3.1, Required: 1.3.1)
[INFO] Checking spdlog (System: 1.15.0, Required: 1.15.0)
[INFO] Checking gtest (Required: 1.17.0)
[INFO] Checking Abseil (System: 20240722, Required: 20240722)
[INFO] Install equivalence dependencies (-eqy)...
[INFO] Checking Yosys (System: 0.58+84, Required: 0.58)
[INFO] Cloning Yosys v0.58..... ✓
[INFO] Building Yosys.....
```

Activities Terminal Oct 24 14:20

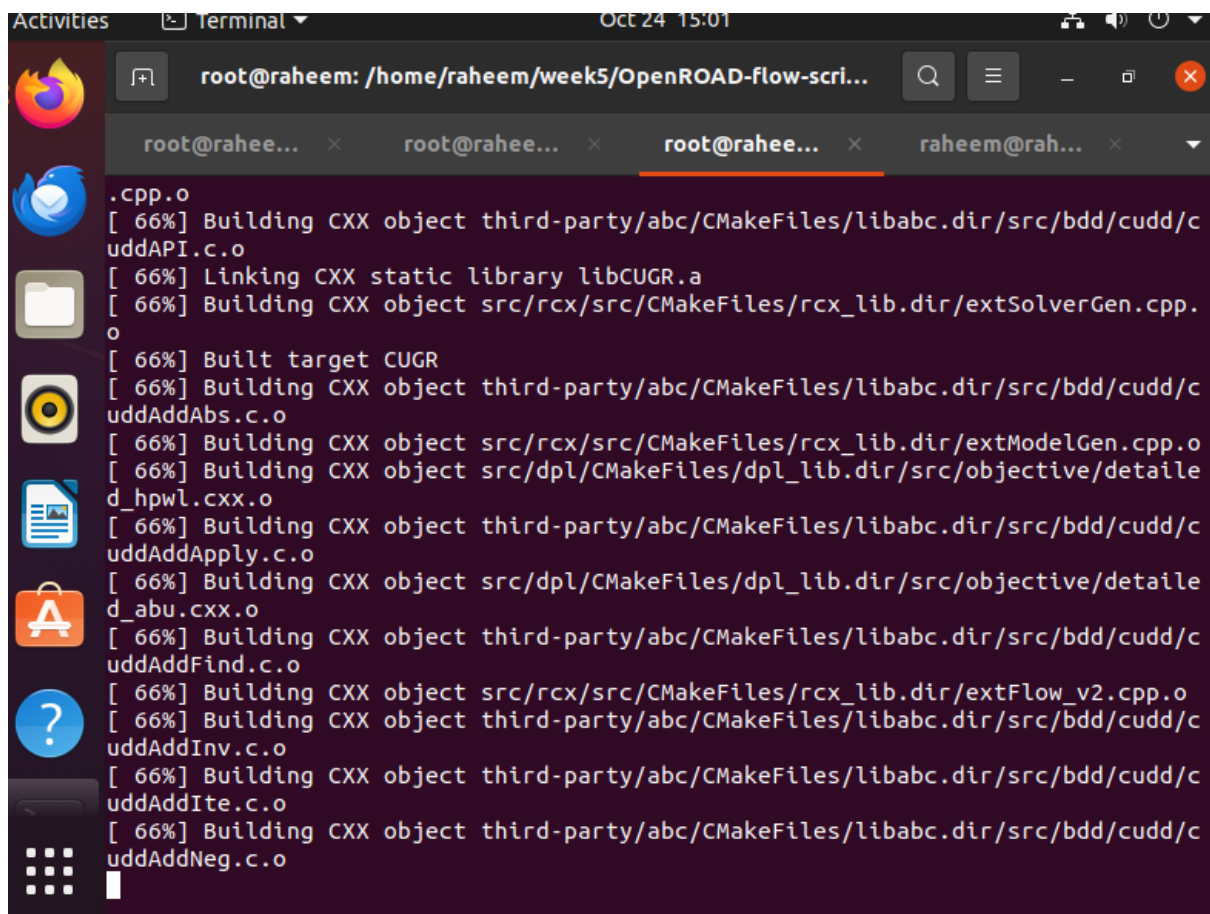
root@raheem: /home/raheem/week5/OpenROAD-flow-scri... root@raheem: /hom... x root@raheem: /hom... x root@raheem: /hom... x

Package	System Version	Required Version	Status
CMake	3.31.9	3.31.9	skipped
Bison	3.8.2	3.8.2	skipped
Flex	2.6.4	2.6.4	skipped
PCRE	10.42	10.42	skipped
SWIG	4.3.0	4.3.0	skipped
Boost	1.86	1.86	skipped
Eigen	3.4	3.4	skipped
CUDD	found	3.0.0	skipped
CUSP	found	any	skipped
Lemon	1.3.1	1.3.1	skipped
spdlog	1.15.0	1.15.0	skipped
gtest	found	1.17.0	skipped
Abseil	20240722	20240722	skipped
Yosys	0.58+84	0.58	installed
Eqy	v0.58	v0.58	skipped
Sby	v0.58-2-g11fd202	v0.58	installed
or-tools	9.11.4210	9.11.4210	skipped

Step 4: Build OpenROAD Locally

Now build OpenROAD itself using the automated build script:

```
./build_openroad.sh --local
```



The screenshot shows a terminal window titled "Terminal" with the date and time "Oct 24 15:01". The terminal is running the command `./build_openroad.sh --local`. The output shows the progress of building various components, with a progress indicator of 66% for each step. The steps include building CXX objects for various libraries and linking static libraries. The terminal window has a dark background and a light-colored text. The window title bar shows the user "root@raheem" and the current directory `/home/raheem/week5/OpenROAD-flow-scri...`. The terminal window is part of a desktop environment with a sidebar on the left containing icons for various applications like Firefox, Thunderbird, and a file manager.

```
root@raheem: /home/raheem/week5/OpenROAD-flow-scri...
root@rahee... x root@rahee... x root@rahee... x raheem@rah... x
.cpp.o
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAPI.c.o
[ 66%] Linking CXX static library libCUGR.a
[ 66%] Building CXX object src/rcx/src/CMakeFiles/rcx_lib.dir/extSolverGen.cpp.
o
[ 66%] Built target CUGR
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAddAbs.c.o
[ 66%] Building CXX object src/rcx/src/CMakeFiles/rcx_lib.dir/extModelGen.cpp.o
[ 66%] Building CXX object src/dpl/CMakeFiles/dpl_lib.dir/src/objective/detaile
d_hpwl.cxx.o
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAddApply.c.o
[ 66%] Building CXX object src/dpl/CMakeFiles/dpl_lib.dir/src/objective/detaile
d_abu.cxx.o
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAddFind.c.o
[ 66%] Building CXX object src/rcx/src/CMakeFiles/rcx_lib.dir/extFlow_v2.cpp.o
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAddInv.c.o
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAddIte.c.o
[ 66%] Building CXX object third-party/abc/CMakeFiles/libabc.dir/src/bdd/cudd/c
uddAddNeg.c.o
```

Some any issues!!! Thank you for VSD Team for this RSIC V Tape-out Program Dr M A Raheem MJCET Hyderabad